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大學生暑期計劃研究成果

AI-Driven De Novo Design of DNA-Binding Proteins

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Objective

The objective of this study was to design a de novo DNA-binding protein derived from the structural features of 3KMD and to evaluate whether the designed protein could adopt a reasonable DNA-binding interface similar to the original TP53-DNA complex.

Method

An integrated computational pipeline was established for DNA-binding protein design. RFD3 was used to generate protein backbone structures around double-stranded DNA, followed by LigandMPNN for amino acid sequence design with DNA as the structural context. Boltz was then used to predict and rank protein–DNA complexes. Finally, ChimeraX was used for structural visualization, interface inspection, hydrogen-bond analysis, and figure preparation.

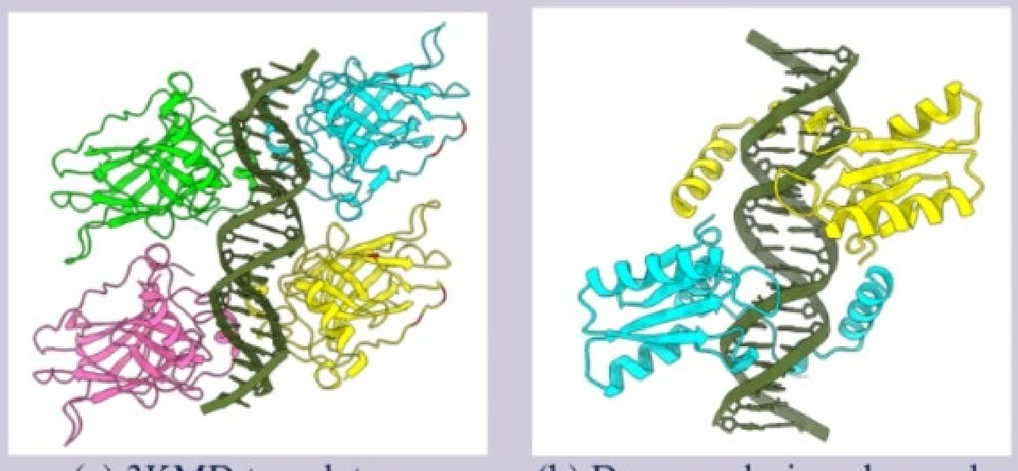
3KMD template > RFD3 > LigandMPNN > Boltz > ChimeraX analysis

Figure 1. AI-assisted computational workflow for de novo DNA-binding protein design.

Results

A series of candidate de novo DNA-binding proteins were generated and screened based on structural prediction and interface analysis. Among the predicted models, the selected candidate, id143, showed the most promising overall structural organization and DNA-binding pose. Compared with the original 3KMD complex, the selected de novo design exhibited a simplified oligomeric organization, a compact protein–DNA interface, and several plausible residue-DNA contacts.

Figure 2. Structural comparison between the 3KMD template and the de novo designed DNA-binding protein.



(a) 3KMD template (b) De novo designed complex

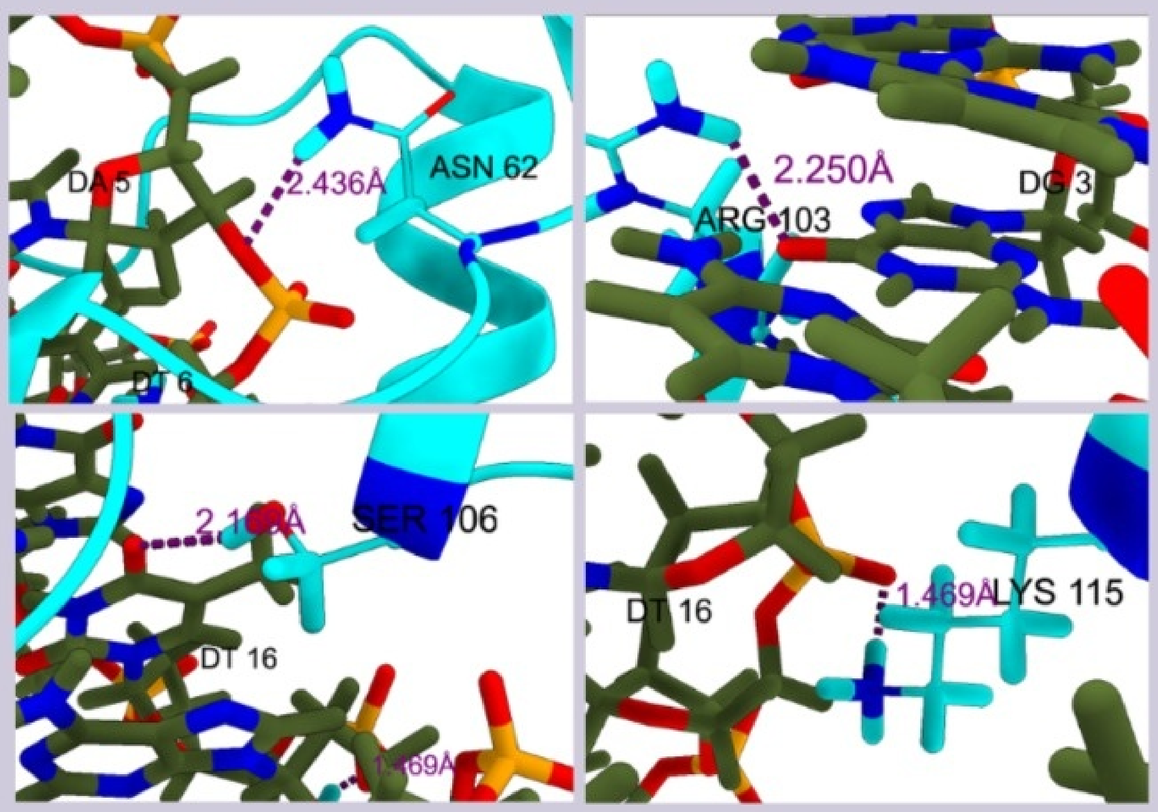
Table 1. Boltz prediction metrics of the selected de novo DNA-binding protein.

Candidate	Confidence score	ipTM	pTM	Complex pLDDT
id143	0.7214	0.8977	0.9003	0.6773

Table 2. Representative residue–DNA contacts identified in the selected model.

Protein residue	DNA target	Contact type	Distance (Å)
ASN62	Adenine 5	Base-associated polar contact	2.436
ARG103	Guanine 3	Base-associated polar contact	2.250
SER106	Thymine 16	Base-associated polar contact	2.169
LYS115	Thymine 16 / phosphate-associated region	Backbone-associated contact	1.469

Figure 3. Representative protein–DNA interactions in the selected de novo design.



Representative protein–DNA contacts were identified in the selected model, including ASN62, ARG103, SER106, and LYS115. ARG103 was positioned near guanine, suggesting a potential base-recognition feature, whereas LYS115 may contribute to DNA backbone stabilization.

Conclusion

This study established an integrated AI-assisted workflow for the de novo design of a 3KMD-derived DNA-binding protein. Using RFD3, LigandMPNN, Boltz, and ChimeraX, we successfully generated and analyzed candidate protein-DNA complexes. The selected model, id143, exhibited a compact DNA-binding architecture and several plausible residue-DNA contacts, indicating that it may serve as a promising candidate for further computational refinement and future experimental validation.